EDET 735 Screen Reader Comparison

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Agenda

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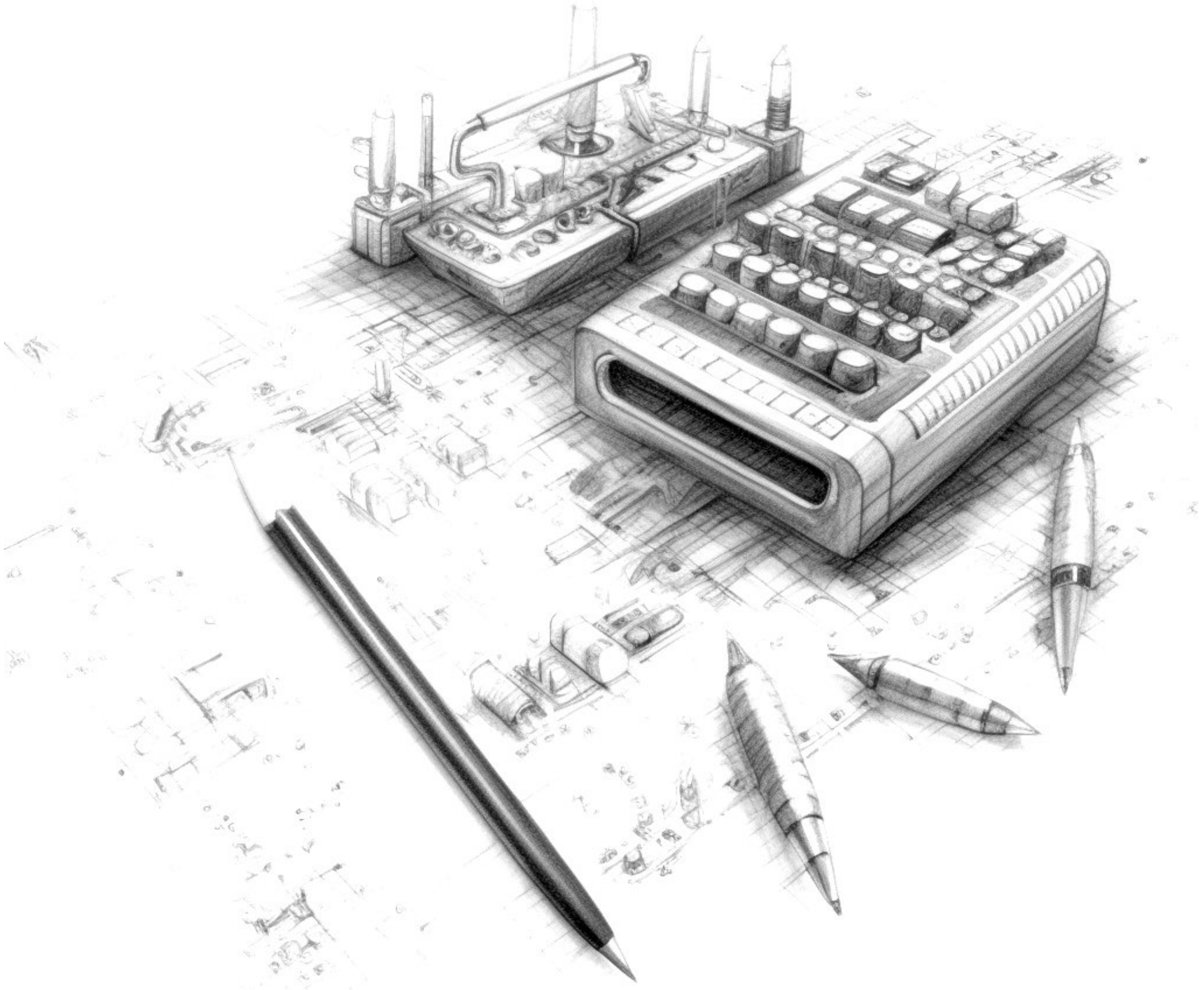
Introductions

Course Overview

- Canvas
- Replit

Module 1

- **Business Problems**
- **Requirements Engineering**
  - Key Terms
  - Techniques
- **Product Requirements Document**
  - Overview
  - Components
- **Tasks**



Course Overview

Learning Management System (LMS)

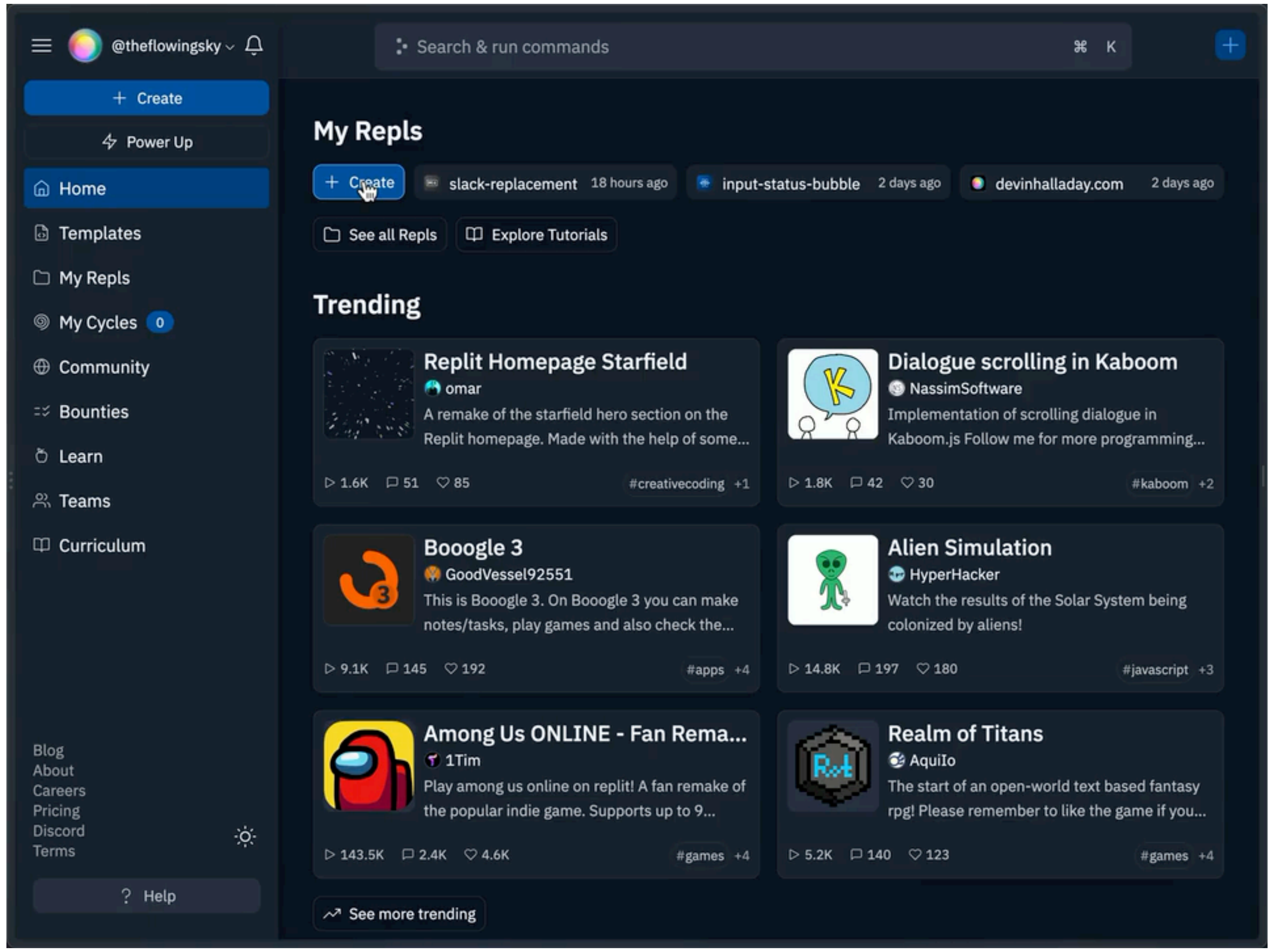
- Facilitates all content, homework, discussions, assignments, and projects
- Canvas MGT 7316 C Course Website

CANVAS
BY INSTRUCTURE

Online Integrated Development Environment (IDE)

- Online code repository which allows development using (m)any programming languages
- Version controlled (integrates with GitHub)
- https://replit.com

replit



Business Problem Statements

Problem Components and Solving Techniques

Problem Statement - a concise description of an issue to be addressed or a condition to be improved upon (includes ideal state, reality, consequences, and proposed solution(s))

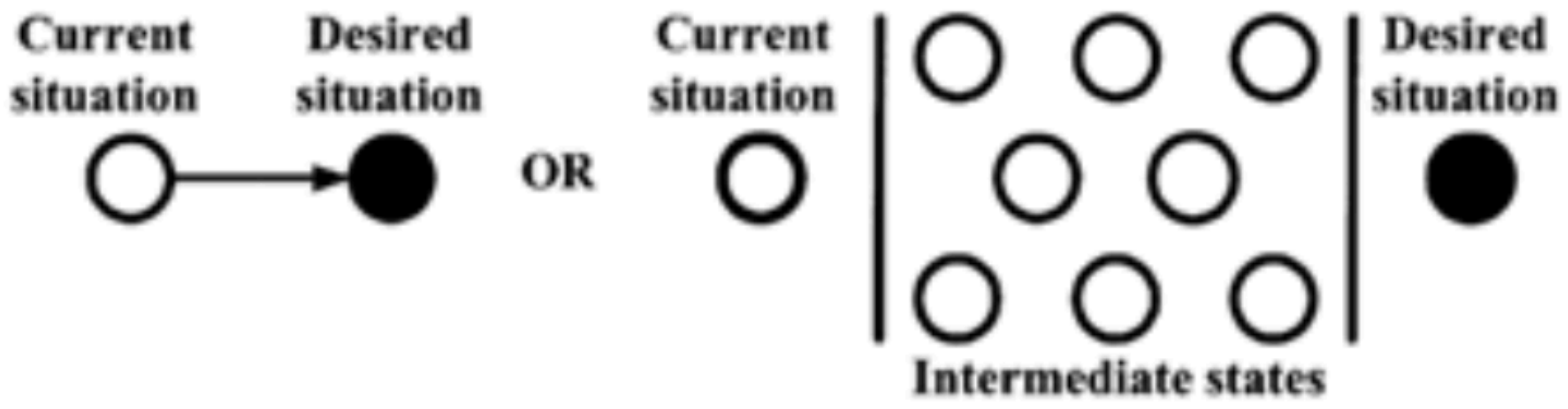


Figure 1: Initial state of Problem solving

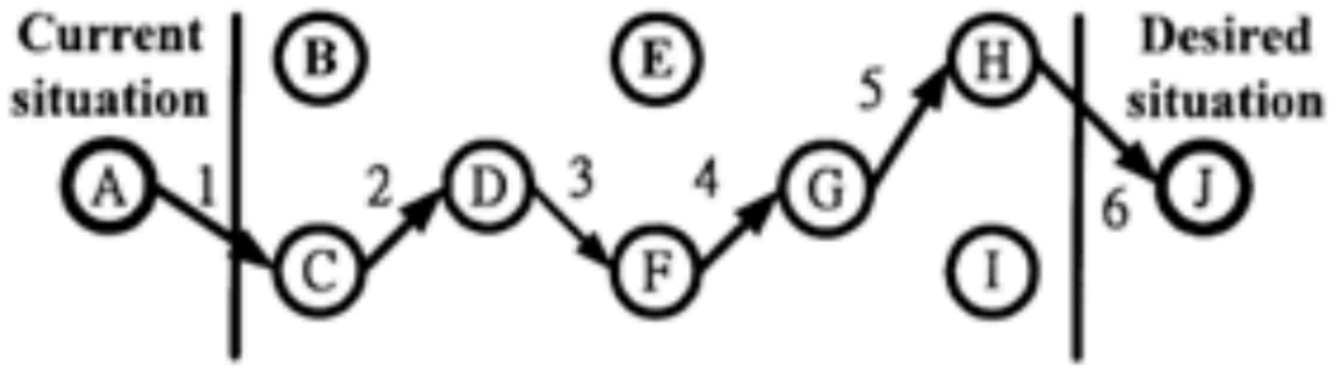


Figure 2: Problem Solving with a chain of intermediate states

Intermediate States - examples include strategic thinking, critical thinking, design thinking, and computational thinking
Strategic Thinking - mental or thinking process applied by an individual in the context of achieving a goal or set of goals
Critical Thinking - analysis of available facts, evidence, observations, and arguments to form a judgment
Design Thinking - an iterative process that teams use to understand users, challenge assumptions, redefine problems and create innovative solutions to prototype and test

Source: https://en.wikipedia.org/wiki/Problem_statement
Source: Importance of Problem Statement in Solving Industry Problems, doi:10.4028/www.scientific.net/AMM.421.857
Source: https://en.wikipedia.org/wiki/Strategic_thinking
Source: https://en.wikipedia.org/wiki/Critical_thinking
Source: https://en.wikipedia.org/wiki/Design_thinking#As_a_process_for_innovation
Source: https://www.interaction-design.org/literature/topics/design-thinking

Key Terms

Coding Syntax

requirement - a singular documented physical or functional need that a particular design, product or process aims to satisfy
functional requirements - defines a function where a function is described as a specification of behavior between inputs and outputs
non-functional requirements (nfr) - criteria used to judge the operation of a system, rather than specific behaviors
(software) engineering - design process to design, develop, maintain, test, and evaluate computer software
requirement engineering (re) - process of defining, documenting, and maintaining requirements in the engineering design process
software development life cycle (SDLC) - process for planning, creating, testing, and deploying an information system

Source: https://en.wikipedia.org/wiki/Requirement
Source: https://en.wikipedia.org/wiki/Functional_requirement
Source: https://en.wikipedia.org/wiki/Non-functional_requirement
Source: https://en.wikipedia.org/wiki/Software_engineering
Source: https://en.wikipedia.org/wiki/Requirements_engineering
Source: https://en.wikipedia.org/wiki/Systems_development_life_cycle

Key Terms

Product Requirements Document (PRD) Structure

title - project name
change history - log of what, who, when the document was changed
overview - brief description
acceptance criteria - what are the success metrics (definition of done)
personas - what are the key identities and/or characteristics of the intended audience
release timeline - project timeline and schedule
functional & non-functional requirements - prioritized feature set
user stories - an event-based script that the browser runs in the
design considerations - references to sketches, style guidelines, branding, etc.
other considerations / axioms / assumptions - catch all for any other items or factors in decision-making

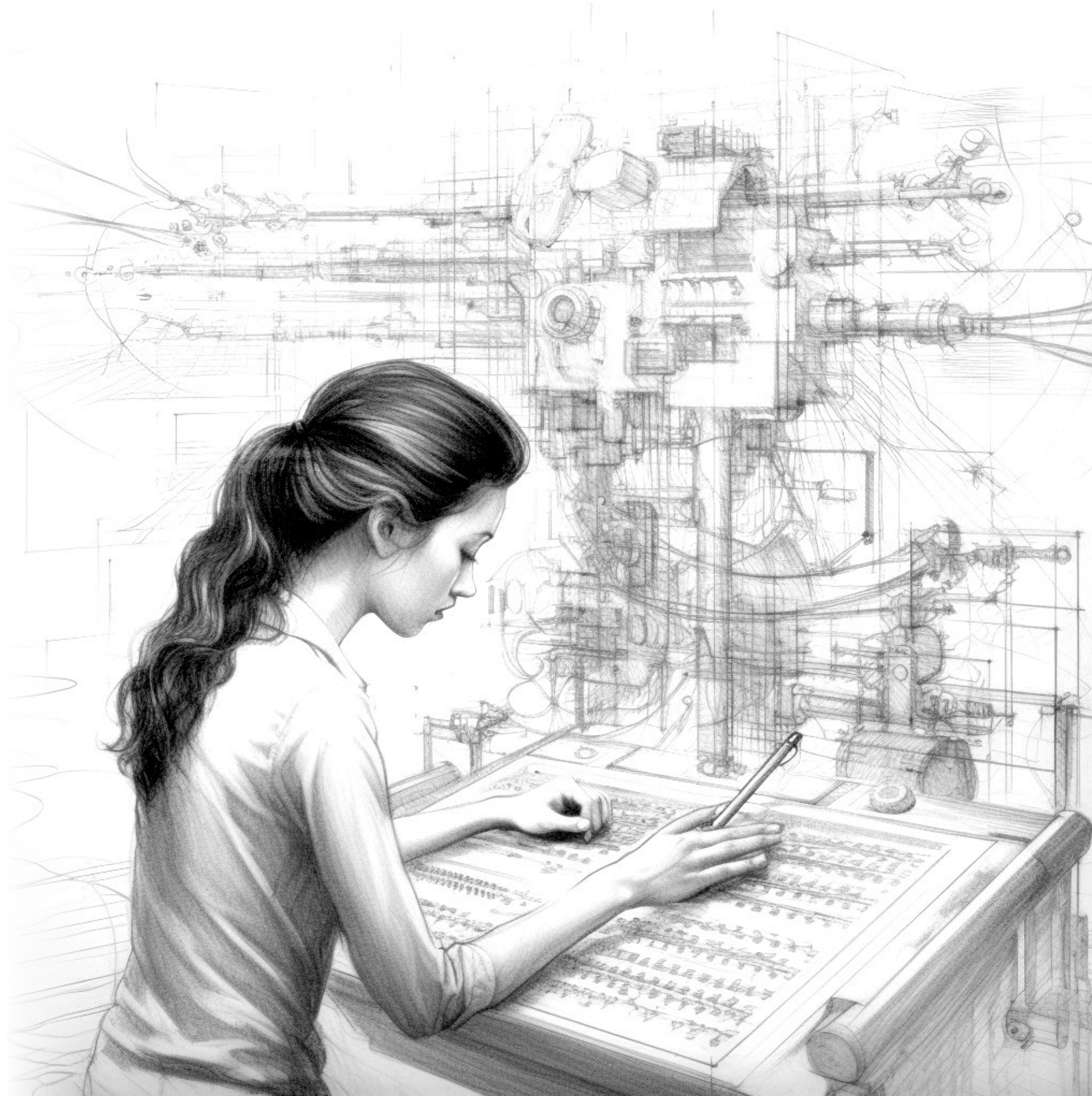
Source: https://productschool.com/blog/product-strategy/product-template-requirements-document-prd

Key Terms

Problem Solving Methodology

1. Create a **problem statement**
2. Use computational thinking as the process which **produces an algorithm/pseudocode** (plain language description of steps) which generates the desired condition or result
3. Convert algorithms/pseudocode (**aka encode)**

In-class exercise: https://replit.com/@johnmcswain/MGT7316PRD



Source: https://en.wikipedia.org/wiki/Pseudocode
Source: https://rogerlmartin.com/lets-read/the-design-of-business

Module 1 Tasks

Readings

Product Requirements Docs (Atlassian)
Google Technical Writing One
Markdown Guide

Module 1 Discussion

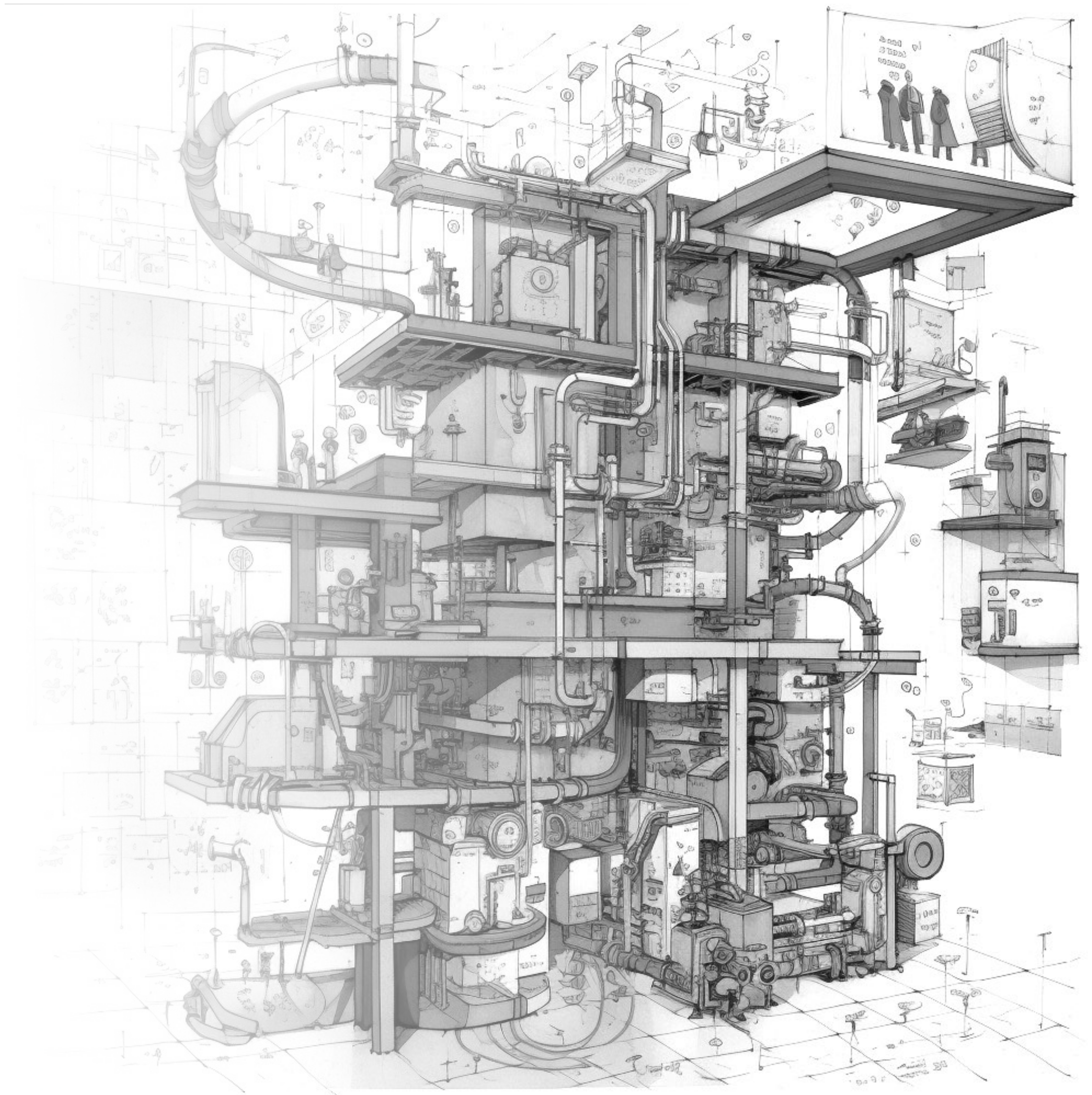
https://canvas.instructure.com/courses/7100862/discussion_topics/18617763

Module 1 Assignment

https://canvas.instructure.com/courses/7100862/assignments/38364328

Pre-assessment Quiz

https://canvas.instructure.com/courses/7100862/assignments/38364328



Questions?